Oluwaseun A. Adekoya

Robotics, Controls, and Automation Engineer | 6+ Years of Experience
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EDUCATION

Ph.D., Mechanical Engineering, University of Cincinnati, OH *Expected Dec 2027*
Dissertation: Safety-Aware Motion Planning and Control for Human–Robot Collaboration: From Industrial Manipulators to Mobile Teleoperated Robots
M.S., Mechanical Engineering, University of Cincinnati, OH *Dec 2024*
B.Eng., Mechanical Engineering, FUTA, Nigeria (First Class, 4.73/5.00; Best Graduating Student) *Nov 2018*
Internship availability: May 2027 – August 2027

RESEARCH EXPERIENCE

Center for Intelligent Metrology and Sensing, University of Cincinnati | Industry Research Engineer *Mar 2026 – Present*
Funded project: Human-Robot Interaction Safety in Manufacturing Using Vision and Machine Learning Technologies (Ohio BWC)

- Developed a human-aware motion planner that achieves over 98.5% speed-and-separation compliance on a 6-DOF Yaskawa HC10.
- Accelerated collision checking to 10–11 ms per 50 samples, eliminating the planning bottleneck from collision computation.
- Engineered look-ahead planning that secures safe paths about 6 secs before worker encroachment; validated on the physical robot.
- Co-authored 3 journal papers under review on RGB-D safety monitoring, safety volumes, and vision-guided fastener removal.
- Programmed and operated a 7-DOF Neuromeka Indy RP2 cobot and a humanoid teleoperation platform.

Integrated Vehicle Design Laboratory, University of Cincinnati | Graduate Researcher *Jan 2021 – Aug 2024*

- **Industry 4.0/5.0 grant (2023):** Led design of real-time nonlinear Model Predictive Control (MPC) for a plug-in hybrid powertrain; sponsored by Siemens, Kinetic Vision, Ethicon, John Deere, and Procter & Gamble.
- **UC Office of Research fellowship (2024):** Benchmarked Model Predictive Control (MPC) against dynamic programming (DP), the global optimum in executing the combined optimal-design of a plug-in hybrid electric vehicle (PHEV) powertrain.
- Achieved a fuel cost of $0.027/mile with real-time MPC and $0.004/mile with DP, against $0.032/mile in prior work.
- Reduced solve time to 4.28 min from about 6 min in prior work.

INDUSTRY EXPERIENCE

Procter & Gamble, Mason, OH | Production Systems and Reliability Engineering Intern *Aug 2024 – Dec 2025*

- Developed reliability models of 6 DayQuil and NyQuil production lines in Personal Health Care to pinpoint likely failure points.
- Increased process reliability from 87.1% to 92.3% and sustained it by targeting the highest-risk failure points to reduce downtime.
- Delivered a similar production-system model to support the launch of a new plant in Greensboro, NC.

Mikano International Ltd., Lagos, Nigeria | Associate Lead Engineer *Apr 2019 – Dec 2020*

- Supervised 4 of 8 generator assembly rows (10 units per row, 4 technicians per unit); the floor built about 80 sets every 3 days.
- Executed assembly, overhaul, and maintenance of generator sets from 9 to 2,000 kVA; ran power audits and SOP reviews.

British American Tobacco, Ibadan, Nigeria | Assistant Project Engineer *Aug 2017 – Jan 2018*

- Supervised piping design and installation for two large desiccant dryers and delivered the 2D CAD models needed for commissioning.

SELECTED PROJECTS

- **RL force control for robotic screwdriving:** Integrated impedance control with reinforcement learning; achieved 98.64% success rate.
- **AI fault detection for production-line labelers:** Achieved about 96% accuracy detecting faulty labelers from vibration sensor data.

SKILLS

Robotics and controls: Motion planning (OMPL, MoveIt), collision avoidance, nonlinear MPC, reinforcement learning, impedance control, ROS 2, sensor integration, computer vision
Programming: Python, C++, MATLAB/Simulink, PyTorch, TensorFlow, JAX, CUDA, Linux (Ubuntu), Docker
Simulation and optimization: MuJoCo, Gazebo, Simscape, CasADi, Acados
Automation and process: PLC (Allen-Bradley), LabVIEW, Arduino, 5S, FMEA, OEE, DFM/DFA, Lean Six Sigma Yellow Belt
Design and fabrication: SolidWorks, Siemens NX, Fusion 360, AutoCAD, Inventor, 3D printing, machining, welding

SELECTED PUBLICATIONS

- **Adekoya, O.**, et al. Safety-Aware Collision Avoidance Using Probabilistic Roadmaps and Human-Aware Motion Planners. In preparation.
- Yahia, T., et al. (incl. **Adekoya, O.**). Experimental Evaluation of RGB-D Sensing for Speed and Separation Monitoring in Human-Robot Collaboration Work Cells. *Robotics and Computer-Integrated Manufacturing*, under review.
- **Adekoya, O.**, Alexander-Ramos, M. A Comparative Study Between Dynamic Programming and Model Predictive Control for Closed-Loop Control Systems. *Journal of Mechanical Design*. (Under Review). Full list: Google Scholar.

HONORS

NSBE Scholarship (2026, 3x) | Paige-Lynn Roberts Memorial Scholarship (2025) | University Research Council Fellowship (2024)
UC Graduate Student Engineer of the Month (2024) | NSF Future of Semiconductors Fellowship (2024) | PEWS Fellowship (2024)